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Procedia Computer Science 219 (2023) 512–520

Procedia Computer Science

www.elsevier.com/locate/procedia

CENTERIS – International Conference on ENTERprise Information Systems / ProjMAN – International Conference on Project MANagement / HCist – International Conference on Health and Social Care Information Systems and Technologies 2022

Design and Implementation of a Product Recommendation System with Association and Clustering Algorithms

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Abstract

Product recommendation systems are an important aspect of retailing because of the improved shopping experience provided for customers. Due to the wide range of products offered by retailers, recommendation systems provide an optimal approach for displaying only relevant products to customers by forming associations that exist between products. Still, it is also important to understand the characteristics of customers connected to different product associations. Conventional approaches for product recommendation systems apply association algorithms and unsupervised classification of customers based on product ratings. However, it is not clear what demographic properties of customers are linked to which different product associations. This paper applies a hybrid system of machine learning (ML) association and clustering algorithms to implement a product recommendation system that shows associations that exist in products and unique customer profiles linked to these associations. The method described in this paper is evaluated with a case of a hygiene product retailer in Austria.

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Peer-review under responsibility of the scientific committee of the CENTERIS – International Conference on ENTERprise Information Systems / ProjMAN - International Conference on Project MANagement / HCist - International Conference on Health and Social Care Information Systems and Technologies 2022

Keywords: Recommendation system; machine learning; product associations; customer profiles; online retailing.

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10.1016/j.procs.2023.01.319

1. Introduction

Retailers usually have a wide variety of products that fall into several product groups and categories. Due to these many product options provided by retailers, customers may have problems in identifying suitable products within a short time when shopping [1]. Also, for online retailers, it is not possible to display the entire list of products to the customers when they use their platforms. Hence, it is necessary to logically display only relevant products in an optimized way. A product recommendation system provides a method of identifying and selecting relevant products by using bidirectional communication between the retail customer and the range of product lists in providing effective and personalized product recommendations [2]. Thus, by displaying selected and individually relevant products to customers, the shopping experience improves due to reduced time spent when shopping [3].

To implement a product recommendation system, two main classes of data are commonly used [4]. First is the characteristic information that provides data about product associations like keywords, categorization, etc. or customer profiles such as age, preferences, location, etc. The second is the information about user and item interactions that include data such as customer ratings, product rankings, etc. According to the review of related articles on different implementations of product recommendation systems in [5], there are three types of recommendation systems which are content-based (CB), collaborative filter based (CFB) and hybrid systems (HS). The content-based approach uses characteristic information from the products customers purchase. The collaborative filtering approach uses interactions resulting in the users and product items. The hybrid systems apply both CB and CFB approaches in realizing product recommendation systems. Thus, customer profiling and association of related products are, therefore, the core part of product recommendation systems. This is because it is necessary to understand products that are associated with each other. Furthermore, it is also important to understand the type and demographic characteristics of customers' connection to each of the product association lists.

Previous research [6] proposed a feature-based and product ranking recommendation system for online purchases. They rank products based on online reviews and use customers' demographic information like age. In [7], a recommendation system for fashion both in online and offline shops has been developed using a new collaborative filtering method. Their algorithm combines data related to online clicking with offline purchasing to weight customer preferences. Extreme Gradient Boosting and Encoding approaches are applied in [8], to improve the performance accuracy of product recommendation systems in an online shopping system. They suggest customer products purchase by investigating users' click patterns.

The focus of this paper is to design and implement a hybrid-based recommendation system that provides both product associations and customer characteristics that are connected to such associations. Previous research [3], applied an association rule mining and clustering approach based on product ratings to link customer profiles to product associations. However, the method does not show the demographic representation of customer profiles that are connected to identified associations. This paper addresses this gap by proposing a method that applies a machine learning (ML) association rule algorithm to outline existing associations and a clustering algorithm to identify the demographic properties of customers linked to these specific associations. Thus, the main research question of this paper is how to apply association and clustering algorithms to implement a recommendation system that shows product associations and demographic representation of customer profiles connected to the associations. For separation of concerns, three research questions are derived from the main research question as follows.

- What is the suitable ML modelling process for showing product associations and connected customer profiles?
- What are the product associations that exist by applying the modelling method?
- What customer demographic profiles are connected to product associations by applying the modelling method?

The rest of the sections are summarized as follows. Section 2 presents the background of the paper showing the commonly used ML algorithms for designing product recommendation systems and a case study of retailers used in evaluating the product recommendation system approach developed in this paper. Section 3 answers the first research question by showing the design of the product recommendation system and the comprising components. Section 4 answers the second and third research questions by showing the product associations and customer profiles obtained by applying the developed recommendation system method. Lastly, section 5 presents the discussions and conclusion of this paper.

2. Background

2.1. ML algorithms for recommendation systems

Depending on the type of recommendation system, different ML algorithms are applied in building such systems. Previous research [4] used K-NN and matrix factorization in designing a hybrid filtering for a recommendation system. In the hybrid system presented in [3], apriori association rule and k-means algorithms are used. K-means algorithm is used in clustering product rankings, while apriori algorithm is used in creating product associations. For the approach proposed in this paper, we apply the K-means algorithm in clustering demographic properties of customers into unique profiles. Then, we apply the apriori algorithm in creating product associations. Details of these two algorithms are presented as follows.

2.1.1. Apriori association rule

The apriori is the most common algorithm for discovering high frequency repeating patterns in a dataset [9]. The algorithm uses a 2-step approach in finding the most associated items in a dataset by applying a sequence of joining and pruning steps. The iteration is repeated until the most frequently repeated associations are identified [10]. The following properties such as antecedent(s)/ consequent support, rule support, confidence and lift are used in describing the apriori algorithm.

- Antecedent(s): an item or a group of items selected by system users before recommendations are put forward.
- Consequent: these are items suggested by the system based on initial selections (antecedents) by the user.
- Support: this is represented by the percentage (%) of antecedent items in the whole dataset.
- Rule support: this is the percentage (%) of occurrence of both the antecedents and consequents in the dataset.
- Confidence: this is the percentage (%) of the conditional probability that a particular consequent will exist for a given occurrence of the antecedent item(s).
- Lift: is a ratio representing the likelihood of an item occurring due to its association to antecedents in comparison to the item occurring independently.

2.1.2. K-means clustering

K-means is a common clustering approach for grouping a given number of observations (n) into a fixed number of clusters (k) in such a way that observations are placed closest to the clusters that have the nearest mean [11]. This clustering algorithm has been applied in classifying products and modelling customer behaviour in retail SCM analytics [12, 13]. In our case, we apply K-means in classifying customer demographic profiles linked to product associations. The observations (n) represent a unique order or sales receipt that is connected to a customer that has a set of demographic properties. For the number of clusters (k), three clusters are on each rule in showing the strength of the contained demographic factors connection a clustered association rule. The largest cluster represents a strong connection, while the smaller clusters can represent a weak or no connection.

2.2. Case presentation: Austrian hygiene product retailer

Following [14], a descriptive single case study with embedded units presenting a first use case of a recommender system based on the combination of ML association and clustering algorithms was conducted. In particular, the recommender system was tested with an Austrian business to business (B2B) hygiene product retailer (>1000 employees, >100€ million turnover, >500 products, operating in more than 60 countries). As a B2B retailer, the company focuses mainly on their vast network of sales personnel engaging in one-to-one conversations to foster the acquisition of new customers. Additionally, it also operates an online shop, which can be used by already acquired customers to buy products without interaction with the sales personnel. However, both forms of acquisition suffer from the huge number of products (>500), which are divided into multiple product groups. As such, in a sales pitch, the salesperson refers to standard items which are always sold to specific customer groups (e.g., hotels, restaurants, hospitals, etc.) and uses gut feeling to determine which other products might be of interest to potential customers. In the online shop, previous customers can search for the items they want to reorder, however, no recommender system

is installed and thus no additional products are shown to these customers. It became apparent, the need of the retailer in providing precise product recommendations for new customers directly in the sales pitch, as well as for returning customers in their online shop.

3. Recommendation System Design

3.1. Components and algorithms

There are four main components of the product recommendations system as shown in Fig. 1. These components include two data source components and two ML association and clustering algorithms. The two data sources provide input data such as product sales and demographic datasets to the association and clustering algorithms, respectively. The output data from the association algorithm which is the associations list provides additional input to the clustering algorithm. The output from the clustering algorithm shows customer profiles associated with a given input association. The apriori algorithm is used for the association rule because it's a common algorithm for implementing product recommendation systems [3, 15]. K-means is used for the clustering part because it has also been used in previous literature for unsupervised classification of customer characteristics linked to different product associations [3].

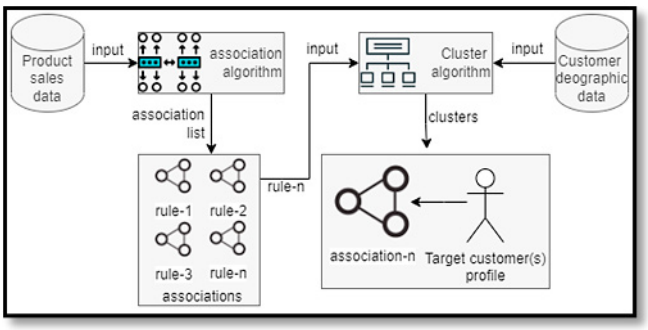

Fig. 1. Design of product recommendation system.

3.2. Recommendation system process

The process capturing the steps for applying the designed product recommendation system is done stepwise as follows: (1) Derive association rules: for a specified minimum support and confidence level, input transactional data containing product sale records are fed into the apriori algorithm to generate a list of product associations. The list shows the support, confidence level, and lift of the association rules generated. (2) Map customers' demographic profiles to associations: for each association rule, a dataset containing basket items of the purchase receipts and their customer demographics are filtered to select records that are contained when the association rule is true. (3) Unsupervised clustering of selected records: the records containing customer demographics when a given association rule is true are clustered with the K-means algorithm to identify the customer profiles that are present for each of the association rules.

4. Results

This section presents the results generated by applying the recommendation system design method in Section 3 to the running case described in Section 2.2. First, the association rules generated using the apriori algorithm are presented. Then, the customer profiles that are connected to the product associations are presented.

4.1. Association Rules

Product associations are generated after applying the association rule algorithm to the data from the running case.

Table 1. Product association rules.

Rule ID	Antecedent 1 Product ID	Antecedent 2 Product ID	Consequent Product ID	Support %	Confidence %	Rule Support %	Lift
8	947	993	13	5.292	79.251	4.194	2.34
9	947	660	13	5.895	79.014	4.658	2.33
1	947	-	13	17.485	77.087	13.478	2.276
3	562	993	660	7.074	65.589	4.781	2.149
6	947	993	660	5.292	63.782	3.376	2.028
4	562	660	993	7.685	62.214	4.781	2.135
10	993	13	660	9.791	62.131	6.083	1.976
2	993	-	660	29.146	57.456	16.746	1.827
7	947	660	993	5.895	57.261	3.376	1.965
11	660	13	993	10.974	55.431	6.083	1.902
5	562	13	660	5.274	55.099	2.906	1.752

Using a configuration of minimum confidence of 55% and minimum support level of 5%, 11 product associations are generated as shown in Table 1 in the order of their confidence level. The first column of the table shows the rule ID, the second and third columns show the 1st and 2nd antecedents respectively, the fourth column shows the percentage of support, the fifth column shows the percentage of the confidence, and the sixth column shows the percentage of the rule support, and the last column shows the lift on each rule ID.

The result of the association rules in the table shows which product(s) is connected to another product with a high level of confidence and which product(s) are connected to the other with the least level of confidence. Also, the association rules with high lift values show that selecting the antecedent product(s) will result in picking the consequent products within the ratios specified as the lift.

4.2. Customer Profiles

The customer profiles of the sales data from the case study are generated by unsupervised clustering of demographic data of customers that made purchases when a given association rule (rule support) is true. The k-means algorithm is used for unsupervised clustering. The demographic variables used for describing the customers are as follows: “designation”, “customer groups”, “state” and “order type”.

Designation: this property defines the attribute of the location of the customers. For instance, some customers are in locations designated as statutory cities, smaller market towns, municipal towns and other city types.

Customer groups: this property provides a group for classifying the type of business establishment the customers are. As already outlined in the case presentation, the type of business dales data is business to business, thus, the customers are represented by business establishments. Nine categories are used in specifying customer groups.

State: this property identifies the geographical state in which the customers are located. There are nine (9) federal states in Austria, and these are used in specifying the location of the customers.

Order type: this property specifies how the customers purchased the products that are contained in each of their sales receipts. The orders could be automated from an ordering device, from the web or manually using sales representatives/ over the telephone. Three categories are used in specifying the customer “order type” property.

Rule-ID: this property is used for filtering sales data and customer demographic data when a given rule is true.

Using a cluster size of 3, Fig. 2 shows the sample of customer profile clusters generated when a given association rule is true (in this case association rule ID = 1). Since cluster size is 3, the maximum customer profiles generated for each rule is three profiles.

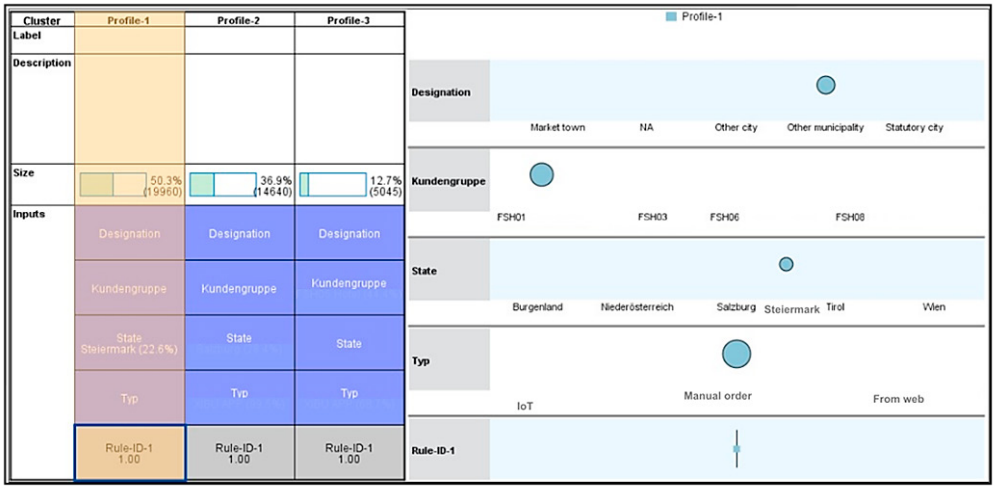

Fig. 2. Sample customer association rule profile clusters.

However, for a profile to be accepted as having a connection to a given rule, the percentage of the sample of sales data in the cluster must be greater than 30%. This condition helps in selecting only clusters with quality connections to the specified association rule.

Table 2. Customer profiles association rules clusters.

No	Rule ID	Profile-ID (Size>30%)	Unique Profile	Desig-nation ID	Customer groups ID	State ID	Order Type ID	Connec-tion
1	1	Profile-1	Profile-A	845	107	358	961	s
2	1	Profile-2	Profile-B	845	107	379	922	w
3	2	Profile-4	Profile-A	845	107	358	961	s
4	3	Profile-7	Profile-C	195	107	358	961	s
5	3	Profile-8	Profile-D	845	671	379	922	w
6	4	Profile-10	Profile-C	195	107	358	961	s
7	4	Profile-11	Profile-D	845	671	379	922	w
8	5	Profile-13	Profile-C	195	107	358	961	s
9	5	Profile-14	Profile-D	845	671	379	922	w
10	6	Profile-16	Profile-A	845	107	358	961	s
11	7	Profile-19	Profile-A	845	107	358	961	s
12	8	Profile-22	Profile-A	845	107	358	961	s
13	9	Profile-25	Profile-A	845	107	358	961	s
14	10	Profile-28	Profile-A	845	107	358	961	s
15	11	Profile-31	Profile-A	845	107	358	961	s
16	11	Profile-32	Profile-D	845	671	379	922	w

Table 2 shows all the profiles generated that satisfy the selection condition. With 11 association rules and 3 cluster sizes, there will be 33 possible customer profiles. By eliminating low cluster samples, only 16 profiles are generated as shown in the table. The first column shows the number of profiles, the second column shows the

association rule, the third column shows the profile ID, and the fourth column shows a unique profile mapping. The next four columns show the demographic properties that are dominant in the profiles. The last column shows the strength of the connection. The unique profile column is used to generate a unique profile for all the 16 selected profiles. This is because the same demographic properties are repeated for some customer profiles. For instance, profile-1, profile-16, profile-19, profile-22, profile-25, profile-28 and profile-31 have the same values of 845, 107, 358 and 961 for demographic factors "designation", "customer groups", "state" and "order type" respectively. Thus, they are labelled as Profile-A. The same applies to the rest of the profiles. Hence, 4 unique customer profiles are generated from Table 2 which are represented as Profile-A, Profile-B, Profile-C and Profile-D. For the profiles connected to the specified association rule, strong (s) and weak (w) is used in outlining the strength of the connection. The profile clusters with the biggest sample size in each rule are assigned the "s" connection, while the profile cluster with the smaller samples is assigned the "w" connection. For instance, Profile-A is strongly connected to rule-ID "1" while Profile-B is weakly connected to the same rule.

4.3. Customer profile association rule clusters

The customer profiles that are connected to the association are generated by mapping the unique profiles to the association rule IDs based on the strength of the connection. Fig. 3 shows the unique customer profiles linked to each of the 11 association rules. The red directed lines from profiles to rules represent the strong connections and the grey directed lines are the weak connections. The association rules are generated in Table 1 while the unique customer profiles are generated in Table 2.

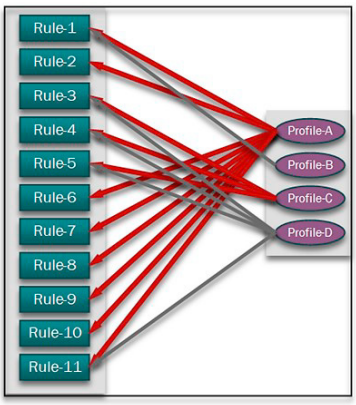

Fig. 3. Customer profile association rule mapping.

5. Discussion, Conclusion and Limitations

To present the scientific discussions emerging from this work, first, we compare the results from this work with similar works, and then, show the managerial implications of this work.

In the recommendation system proposed in [8], for grocery retailing, the system works by customer product ratings and associations linked to the product. For new customers, that haven't picked any product, the system recommends sample products from the list of highly ranked products. Then, the associations are provided to the customers based on the initial choice. However, in the approach described in our paper, new customers can be assigned to a different cluster of customer profiles they belong to. Thus, the product list in all the associations linked to the assigned customer profile can be initially advertised to the customer. Thereafter, depending on the customer's first product choice, product associations connected to the selected product are displayed. The paper [9] shows the use of apriori algorithm in implementation of a product recommendation system for an online store. By using a high support level (>40%) and confidence level (>70%), the result showed only a single association rule with a confidence level of 76%. However, we applied the same Apriori algorithm with low levels of confidence (>55%) and support (>5%), and our result showed three association rules with confidence levels that are over 76%. Thus,

increasing the sample data in consideration by applying a lower support level in the apriori algorithm can result in association rules that have high confidence values. Still, in different industries, the demographic properties used in establishing the customer profiles may vary from the results obtained with our current business case. Thus, additional work in seperate case-studies can be used to further validate the method developed in this paper.

Our research also comes with several managerial implications. As such, we could demonstrate that using a combination of association and clustering algorithms can lead to specific product associations and customer characteristics. Looking at our specific case, the product associations could be used by the hygiene product retailer to improve customer acquisition. Where previously, salespersons relied on a pre-defined list of standard items which were always sold to specific customer groups and their gut feeling to determine which other products might be of interest to potential customers, they are now able to classify potential customers based on previously collected "demographic data" and thus offer them specific items which are, according to sales data, always bought by specific customer groups. This reduces the amount of "guesswork" the sales personnel must do and replaces it with data-driven decision making which according to [16], has the potential to increase sales and customer retention. Similarly, this approach could also be implemented in the retailers' online shops. There it can offer the retailers' returning customers decision support by displaying them additional items they might also want to buy based on their previous purchase and customer profile. Hence, similarly as demonstrated in [17], the designed recommender system can also work as an upselling tool. Overall, our research could thus demonstrate that the combination of association and clustering algorithms can be used by companies to assist their sales personnel and customers in decision making.

The objective of this paper is to implement a product recommendation system that applies a hybrid ML approach in showing product associations and customer demographic profiles linked to the associations. To achieve this, we designed a recommendation system that combines inputs from sales data and customer demographic data using the apriori association rule and K-means clustering algorithms. The results show eleven (11) groups of products associated with each other and the level of confidence and lift ratios of the identified associations. The result furthermore shows the four unique customer profiles that are strongly/ weakly linked to the identified associations.

Every research has some limitations, and so does our paper. First, the presented results are based on a single case study only and thus might not be representative of other retailers from other industries. Second, we were only supplied with a limited amount of demographical data to perform our customer clustering, which might have reduced the overall quality of the customer clusters. Lastly, although our recommender system shows good performance in terms of showing product associations and clustering demographics linked to the associations, it has not yet been implemented in an actual retailer sales platform. Thus, is it not possible to evaluate the impact on a real business. Hence, the future work resulting from this research is to evaluate the business impact of the implementation of our recommendation system. Additionally, future studies could work on more advanced demographic data to further increase the clustering accuracy. Also, the implementation of a recommendation system based on association and clustering algorithms should be tested in other industries to ensure its general applicability.

Acknowledgements

This research work was funded by the Government of Upper Austria as part of the Logistikum.RETAIL.

References

[1] D. Burch and G. Lawrence, "Supermarket Own Brands, Supply Chains and the Transformation of the Agri-food System," 2005, doi: 10.48416/IJSAF.V13I1.312.

[2] S. H. Choi, S. Kang, and Y. J. Jeon, "Personalized recommendation system based on product specification values," Expert systems with applications, vol. 31, no. 3, pp. 607–616, 2006, doi: 10.1016/j.eswa.2005.09.074.

[3] P. Satheesan, P. S. Haddela, and J. Alosius, "Product Recommendation System for Supermarket," in 2020 19th IEEE International Conference on Machine Learning and Applications (ICMLA), Miami, FL, USA, 2020, pp. 930–935.

[4] Jatin Sharma, Kartikay Sharma, Kaustubh Garg, and Avinash Kumar Sharma, "Product Recommendation System a Comprehensive Review," IOP Conference Series: Materials Science and Engineering, vol. 1022, no. 1, p. 12021, 2021, doi: 10.1088/1757-899x/1022/1/012021.

[5] P. Kumar and R. S. Thakur, "Recommendation system techniques and related issues: a survey," Int. j. inf. tecnol., vol. 10, no. 4, pp. 495–501, 2018, doi: 10.1007/s41870-018-0138-8.

[6] R. V. Karthik, S. Ganapathy, and A. Kannan, “A Recommendation System for Online Purchase Using Feature and Product Ranking,” in 2018 Eleventh International Conference on Contemporary Computing (IC3), Noida, 2018, pp. 1–6.

[7] H. Hwangbo, Y. S. Kim, and K. J. Cha, “Recommendation system development for fashion retail e-commerce,” Electronic Commerce Research and Applications, vol. 28, pp. 94–101, 2018, doi: 10.1016/j.elerap.2018.01.012.

[8] Z. Shahbazi, D. Hazra, S. Park, and Y. C. Byun, “Toward Improving the Prediction Accuracy of Product Recommendation System Using Extreme Gradient Boosting and Encoding Approaches,” Symmetry, vol. 12, no. 9, p. 1566, 2020, doi: 10.3390/sym12091566.

[9] P. Edastama, A. S. Bist, and A. Prambudi, “Implementation Of Data Mining On Glasses Sales Using The Apriori Algorithm,” IJCITSM, vol. 1, no. 2, pp. 159–172, 2021, doi: 10.34306/ijcitsm.v1i2.46.

[10] Jai Puneet Singh and Hari Ram, “Improving Efficiency of Apriori Algorithm Using Transaction Reduction,” in 2013.

[11] A. Likas, N. Vlassis, and J. J. Verbeek, “The global k-means clustering algorithm,” Pattern Recognition, vol. 36, no. 2, pp. 451–461, 2003, doi: 10.1016/S0031-3203(02)00060-2.

[12] P. Anitha and M. M. Patil, “RFM model for customer purchase behavior using K-Means algorithm,” Journal of King Saud University - Computer and Information Sciences, vol. 34, no. 5, pp. 1785–1792, 2022, doi: 10.1016/j.jksuci.2019.12.011.

[13] K. Kusrini, “Grouping of Retail Items by Using K-Means Clustering,” Procedia Computer Science, vol. 72, pp. 495–502, 2015, doi: 10.1016/j.procs.2015.12.131.

[14] R. K. Yin, Case study research: Design and methods, 4th ed. Thousand Oaks: Sage, 2009.

[15] C. S. Fatoni, E. Utami, and F. Wibowo, “Online Store Product Recommendation System Uses Apriori Method,” Journal of Physics: Conference Series, vol. 1140, p. 12034, 2018, doi: 10.1088/1742-6596/1140/1/012034.

[16] Amatriain, Xavier, and Justin Basilico. "Recommender systems in industry: A netflix case study." Recommender systems handbook. Springer, Boston, MA, 2015. 385-419.

[17] Ganesh, Jaishankar, et al. "Online shopper motivations, and e-store attributes: an examination of online patronage behavior and shopper typologies." Journal of retailing 86.1 (2010): 106-115.